

Express the product $A\mathbf{x}$ as a linear combination of the column vectors of A .

2.

(a) $\begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(b) $\begin{bmatrix} 4 & 0 & -1 \\ 3 & 6 & 2 \\ 0 & -1 & 4 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \\ 5 \end{bmatrix}$

Question 02:

Part (a)

(a)

$$\begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \cdot 1 + 3 \cdot 2 \\ (-1) \cdot 1 + 4 \cdot 2 \end{bmatrix} = 1 \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Part (b)

(b)

$$\begin{bmatrix} 4 & 0 & -1 \\ 3 & 6 & 2 \\ 0 & -1 & 4 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \cdot (-2) + 0 \cdot 3 + (-1) \cdot 5 \\ 3 \cdot (-2) + 6 \cdot 3 + 2 \cdot 5 \\ 0 \cdot (-2) + (-1) \cdot 3 + 4 \cdot 5 \end{bmatrix} = (-2) \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 6 \\ -1 \end{bmatrix} + 5 \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}$$

Find a basis for the nullspace of A .

6.

$$(a) \quad A = \begin{bmatrix} 1 & -1 & 3 \\ 5 & -4 & -4 \\ 7 & -6 & 2 \end{bmatrix}$$

(a)

Reducing matrix

$$\begin{bmatrix} 1 & -1 & 3 \\ 5 & -4 & -4 \\ 7 & -6 & 2 \end{bmatrix} \xrightarrow{(1)} \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -19 \\ 0 & 1 & -19 \end{bmatrix} \xrightarrow{(2)} \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -19 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{(3)} \begin{bmatrix} 1 & 0 & -16 \\ 0 & 1 & -19 \\ 0 & 0 & 0 \end{bmatrix}$$

Explanations:

- (1) Multiply row 1 by -5 and add to row 2. Multiply row 1 by -7 and add to row 3.
- (2) Subtract row 2 from row 3.
- (3) Add row 2 to row 1.

So if

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

is solution of the system $A\mathbf{x} = \mathbf{0}$ then

$$x_1 - 16x_3 = 0$$

$$x_2 - 19x_3 = 0$$

If we set $x_3 = t$ then $x_1 = 16t$, $x_2 = 19t$ and

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = t \begin{bmatrix} 16 \\ 19 \\ 1 \end{bmatrix}$$

Therefore

$$\left\{ \begin{bmatrix} 16 \\ 19 \\ 1 \end{bmatrix} \right\}$$

is basis for the null space of A .

$$(c) \quad A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix}$$

Reducing matrix

$$\begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix} \xrightarrow{(1)} \begin{bmatrix} 1 & 4 & 5 & 2 \\ 0 & -7 & -7 & -4 \\ 0 & 7 & 7 & 4 \end{bmatrix} \xrightarrow{(2)} \begin{bmatrix} 1 & 4 & 5 & 2 \\ 0 & -7 & -7 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{(3)} \begin{bmatrix} 2 & 1 & 3 & 0 \\ 0 & -7 & -7 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$A\mathbf{x} = \mathbf{0}$ then

$$\begin{aligned} 2x_1 + x_2 + 3x_3 &= 0 \\ -7x_2 - 7x_3 - 4x_4 &= 0 \end{aligned}$$

If we set $x_2 = t$, $x_3 = s$ then $x_1 = -\frac{1}{2}t - \frac{3}{2}s$, $x_4 = -\frac{7}{4}t - \frac{7}{4}s$ and

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}t - \frac{3}{2}s \\ t \\ s \\ -\frac{7}{4}t - \frac{7}{4}s \end{bmatrix} = t \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \\ -\frac{7}{4} \end{bmatrix} + s \begin{bmatrix} -\frac{3}{2} \\ 0 \\ 1 \\ -\frac{7}{4} \end{bmatrix}$$

Therefore

$$\left\{ \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \\ -\frac{7}{4} \end{bmatrix}, \begin{bmatrix} -\frac{3}{2} \\ 0 \\ 1 \\ -\frac{7}{4} \end{bmatrix} \right\}$$

is basis for the null space of A .

13. Prove that the row vectors of an $n \times n$ invertible matrix A form a basis for $\mathbb{R}^n$.

Let A be an $n \times n$ invertible matrix. Since A^T is also invertible, it is row equivalent to I_n . It is clear that the column vectors of I_n are linearly independent. Hence, by virtue of Theorem 5.5.5, the column vectors of A^T , which are just the row vectors of A , are also linearly independent. Therefore the rows of A form a set of n linearly independent vectors in $\mathbb{R}^n$, and consequently form a basis for $\mathbb{R}^n$.

THEOREM 5.5.5

If A and B are row equivalent matrices, then

- (a) *A given set of column vectors of A is linearly independent if and only if the corresponding column vectors of B are linearly independent.*
- (b) *A given set of column vectors of A forms a basis for the column space of A if and only if the corresponding column vectors of B form a basis for the column space of B .*

THEOREM 5.5.4

Elementary row operations do not change the row space of a matrix.

Proof Suppose that the row vectors of a matrix A are $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_m$, and let B be obtained from A by performing an elementary row operation. We shall show that every vector in the row space of B is also in the row space of A and that, conversely, every vector in the row space of A is in the row space of B . We can then conclude that A and B have the same row space.

Consider the possibilities: If the row operation is a row interchange, then B and A have the same row vectors and consequently have the same row space. If the row operation is multiplication of a row by a nonzero scalar or the addition of a multiple of one row to another, then the row vectors $\mathbf{r}'_1, \mathbf{r}'_2, \dots, \mathbf{r}'_m$ of B are linear combinations of $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_m$; thus they lie in the row space of A . Since a vector space is closed under addition and scalar multiplication, all linear combinations of $\mathbf{r}'_1, \mathbf{r}'_2, \dots, \mathbf{r}'_m$ will also lie in the row space of A . Therefore, each vector in the row space of B is in the row space of A .

Since B is obtained from A by performing a row operation, A can be obtained from B by performing the inverse operation (Section 1.5). Thus the argument above shows that the row space of A is contained in the row space of B . ■

In light of Theorems Theorem 5.5.3 and Theorem 5.5.4, one might anticipate that elementary row operations should not change the column space of a matrix. However, this is *not* so— elementary row operations can change the column space. For example, consider the matrix

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$$

The second column is a scalar multiple of the first, so the column space of A consists of all scalar multiples of the first column vector. However, if we add -2 times the first row of A to the second row, we obtain

$$B = \begin{bmatrix} 1 & 3 \\ 0 & 0 \end{bmatrix}$$

Here again the second column is a scalar multiple of the first, so the column space of B consists of all scalar multiples of the first column vector. This is not the same as the column space of A .

Although elementary row operations can change the column space of a matrix, we shall show that whatever relationships of linear independence or linear dependence exist among the column vectors prior to a row operation will also hold for the corresponding columns of the matrix that results from that operation. To make this more precise, suppose a matrix B results from performing an elementary row operation on an $m \times n$ matrix A . By Theorem 5.5.3, the two homogeneous linear systems

$$A\mathbf{x} = \mathbf{0} \quad \text{and} \quad B\mathbf{x} = \mathbf{0}$$

have the same solution set. Thus the first system has a nontrivial solution if and only if the same is true of the second. But if the column vectors of A and B , respectively, are

$$\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n \quad \text{and} \quad \mathbf{c}'_1, \mathbf{c}'_2, \dots, \mathbf{c}'_n$$

then from 2 the two systems can be rewritten as

$$x_1\mathbf{c}_1 + x_2\mathbf{c}_2 + \dots + x_n\mathbf{c}_n = \mathbf{0} \tag{6}$$

and

$$x_1\mathbf{c}'_1 + x_2\mathbf{c}'_2 + \dots + x_n\mathbf{c}'_n = \mathbf{0} \tag{7}$$

Thus 6 has a nontrivial solution for $x_1, x_2, \dots, x_n$ if and only if the same is true of 7. This implies that the column vectors of A are linearly independent if and only if the same is true of B . Although we shall omit the proof, this conclusion also applies to any subset of the column vectors. Thus we have the following result.